

## Pipeline Description

The pipeline structure follows the Snakemake modularization. Modules are set in a main snake file which will be called for the workflow execution (Figure 1). A module consists of one or more rules which decompose the workflow into small steps by specifying how to create sets of output files from sets of input files. A rule can not be run if the input files are not found, also if an error occurs during the execution the output files created are deleted to keep consistency in the workflow in case a corrupted output is created. A rule also could include configurable parameters such as params, shell, messages, among others (Figure 2). File names, paths and parameters can be set in specific configuration files for easy modifications and administration (Figure 3 and Figure 4).

Most of the pipeline dependencies are managed using Conda through the Snakemake Integrated Package Management. File dependenciesenv.yml (env directory) should be modified to update any dependency (Figure 5).

```
# Config files containing hard coded variables
configfile: os.path.abspath(os.path.join(os.path.dirname(workflow.basedir), 'snakefiles/config.json'))

# Runtime variables
include: "variables.py"

# rule all
rule all:
    input:
        dupconcordant = config['data_dup_path'] + '/' + config['dup_concordant_file'],
        dupdiscordant = config['data_dup_path'] + '/' + config['dup_disconcordant_file']
        samplespassmerged = config['sample_merged_file']

# Include modules
include: "module1-data-conversion.snake"
include: "module2-data-calling.snake"
include: "module3-data-clean.snake"
```

**Figure 1.** Snakemake file. This file includes the main rule “all” (by default snakemake executes this rule), and all the modules which will be executed in the pipeline workflow. In this file you can define your pipeline adding or removing any module allowing a flexible workflow. The file config.json contains the paths names to access the files described in each rule. And, the variable.py file contains the path that should be created for the workflow structure results. This file also can define links to external tools, libraries and resources which could be called in any rule. And, it also can define global parameters and external parameters for external tools.

```

# Run PennCNV to detect CNVs
# 1. Input: PFB and GCmodel files
# 2. output: list of calls by sample in *.rawcn file
rule DETECT_CNVs:
    input:
        pbffile = config['pfb_file'],
        gcmodelfile = config['gcmodel_file']
    output:
        call_raw = expand(config['data_calling_path'] + '/' + '{prefix}' + '.rawcn', prefix=calling_prefix),
        call_log = expand(config['data_calling_path'] + '/' + '{prefix}' + '.log', prefix=calling_prefix)
    params:
        signalintensitydir = config['data_intensity_path'],
        outputdir = config['data_calling_path'],
        hmmfile = config['hmm_file'],
        listfile = config['list_signal_files_file'],
        logfile = config['log_path'] + '/' + 'data_calling.log',
    message: "Calling for CNV detection from {input} in {output}"

    shell:
        """
        #Calling from PennCNV command
        detect_cnv.pl \
            -test \
            -conf \
            -hmm {params.hmmfile} \
            -pfb {input.pbffile} \
            -directory {params.signalintensitydir} \
            -list {params.listfile} \
            -log {output.call_log} \
            -out {output.call_raw} \
            -gcmodel {input.gcmodelfile}

        #login
        samples_with_calls=`awk '{{print $5}}' {output.call_raw} | sort -u | wc -l`
        total_calls=`wc -l {output.call_raw} | cut -f1 -d' '`
        echo "Were found $samples_with_calls with $total_calls calls" >> {params.logfile}
        """

```

**Figure 2.** Module and rule structure. This figure shows a Module composed by one rule. Most commonly, rules consist of a name, input files, output files, and a shell command to generate the output from the input. This rule also contains the *params* specification which defines the alternative parameters used in this rule. Labels in red specify the path name for a file or directory. They are defined in the config.json file.

```

{
    "final_report_file": "/data/GSA2016_308_025_FinalReport.txt",
    "signal_intensity_file": "/data/signal_intensity.txt",

    "gc_content_file": "RareCNVsAnalysis/qc-cnv/resources/gc5Base.sorted.txt",
    "hmm_file": "RareCNVsAnalysis/qc-cnv/resources/hhall.hmm",
    "immunoglobulin_region_file": "RareCNVsAnalysis/qc-cnv/resources/immunoglobulin_penncnv.txt",
    "centromere_telomere_region_file": "RareCNVsAnalysis/qc-cnv/resources/centromere_telomere_penncnv.txt",

    "list_signal_files_file": "QCrResults/data_conversion/list.txt",
    "map_file": "QCrResults/data_conversion/sample_map.txt",
    "snp_file": "QCrResults/data_conversion/SNPfile.txt",
    "pfb_file": "QCrResults/data_conversion/model.pfb",
    "gcmodel_file": "QCrResults/data_conversion/hg19.gcmodel",
    "sample_pass_list_file": "QCrResults/data_clean/samples_qcpass.list",
    "sample_pass_file": "QCrResults/data_clean/samples_qcpass.rawcn",
    "sample_summary_file": "QCrResults/data_clean/samples_qcsum.list",
    "sample_clean_file": "QCrResults/data_clean/samples_qcpass.clean.rawcn",
    "sample_merged_file": "QCrResults/data_clean/samples_qcpass.clean.merged.rawcn",

    "data_conversion_path": "QCrResults/data_conversion",
    "data_intensity_path": "QCrResults/data_conversion/data_intensity",
    "data_calling_path": "QCrResults/data_calling",
    "data_clean_path": "QCrResults/data_clean",
    "graphic_path": "QCrResults/graphics",
    "graphic_qc_path": "QCrResults/graphics/qc",
    "log_path": "QCrResults/logs",

    "dependenciesenv_file": "RareCNVsAnalysis/qc-cnv/qc-pipeline/snakefiles/env/dependenciesenv.yml"
}

```

**Figure 3.** Config.json file. This file contains file and directory paths for input and output data. Any change in the file location just has to be modified in this file. Similarly, the output directory can be changed just by modifying this file. Changes are broadcast to all modules and rules. Also this file describes the environment dependencies file path.

```

### snakemake_workflows initialization #####
libdir = os.path.abspath(os.path.join(os.path.dirname(workflow.basedir), '../lib'))
resourcesdir = os.path.abspath(os.path.join(os.path.dirname(workflow.basedir), '../resources'))

### programs #####
#Include here all programs and versions.You can run the specific program/version
#calling it as {program_version} inside the code. E.g {R_3_4}
pennCNV = "/home/haydee.artaza/programs/PennCNV-1.0.5"
R_4_1 = "/home/haydee.artaza/programs/R_4_1"
### prefix #####
### module 1,2 and 3
signal_prefix = "split"
calling_prefix = "sampleall"

### Workflow parameters #####
### File extensions
PLINK_EXT = ['.bed', '.bim', '.fam']
TPLINK_EXT = ['.tped', '.tfam']
### PennCNV
qcnumcnv = "50"
wf = "0.05"
qcbafrdrift = "0.01"
qclrrsd = "0.3"

### Create paths if don't exist #####

if not os.path.exists(config['log_path']):
    os.makedirs(config['log_path'])
if not os.path.exists(config['data_conversion_path']):
    os.makedirs(config['data_conversion_path'])
if not os.path.exists(config['data_intensity_path']):
    os.makedirs(config['data_intensity_path'])
if not os.path.exists(config['data_calling_path']):
    os.makedirs(config['data_calling_path'])
if not os.path.exists(config['data_clean_path']):
    os.makedirs(config['data_clean_path'])
if not os.path.exists(config['graphic_path']):
    os.makedirs(config['graphic_path'])
if not os.path.exists(config['graphic_qc_path']):
    os.makedirs(config['graphic_qc_path'])

```

**Figure 4.** The variable.py file is a configuration file which contains the path structure created for the workflow to store results. This file also can define links to external tools, libraries and resources which could be called in any rule. And, it also can define global parameters and external parameters for external tools.

```

channels:
- conda-forge
- defaults
- r
- bioconda
dependencies:
- r=4.2
- r-ggplot2
- r-fmsb
- r-gridExtra
- r-dplyr
- r-reshape
- r-devtools
- openssl
- fontconfig
- harfbuzz
- freibidi
- freetype
- libpng
- libtiff
- libjpeg-turbo
- libcurl
- cmake

```

**Figure 5.** YAML dependencies configuration file. Software (and the optional version) run in any rule should be included in this file.

## Calling and quality control pipeline

### 1. Module Data Conversion

#### Rule CREATE\_INTENSITY\_FILES

**Description:** Create the individual signal intensity files, the list of the signals name per individual, and the list of SNPs for PennCNV calling. Also create the map file containing the original ID sample with the new name assigned by PennCNV (split\* when using the `--numeric` option to hide the original ID for sensitive data).

**Input:** signal intensity and Illumina report (Figure 6 and Figure 7).

**Output:** a file with the list of names for each intensity file created (list.txt), the SNPs list file (name, Chr, Position), and the individual signal intensity (Figure 8).

A log file is generated inside the logs directory with the module name (logs/data\_conversion.log) showing useful information for future reports.

| Name        | Chr | Position  | Patient1.GType | Patient1.B Allele Freq | Patient1.Log R Ratio |
|-------------|-----|-----------|----------------|------------------------|----------------------|
| 1:100292476 | 1   | 100292476 | AA             | 0,00                   | -0,3792592           |
| 1:101064936 | 1   | 101064936 | AA             | 0,01713777             | -0,1650909           |
| 1:103380393 | 1   | 103380393 | BB             | 0,95952                | -0,02254537          |
| 1:104303716 | 1   | 104303716 | BB             | 0,9974257              | -0,09440669          |
| 1:104864464 | 1   | 104864464 | BB             | 1                      | -0,08369645          |

**Figure 6.** SNPs table file format. This figure shows the extraction of one patient intensity signal for the first four markers.

```
GSMT Version      2.0.4
Processing Date    1/17/2020 11:48 AM
Content            GSAMD-24v1-0_20011747_A4.bpm
Num SNPs           700078
Total SNPs         700078
Num Samples        6112
Total Samples      6112
[Data]
SNP Name           Sample ID           B Allele Freq    Log R Ratio
1:100292476        Patient1             0.0000           -0.3793
1:101064936        Patien1             0.0171           -0.1651
1:103380393        Patient1            0.9595           -0.0225
1:104303716        Patient1            0.9974           -0.0944
1:104864464        Patient1            1.0000           -0.0837
```

**Figure 7.** Final report from Genome studio format

| Name        | ID.Log R Ratio | ID.B Allele Freq |
|-------------|----------------|------------------|
| 1:100292476 | -0.3793        | 0.0000           |
| 1:101064936 | -0.1651        | 0.0171           |
| 1:103380393 | -0.0225        | 0.9595           |
| 1:104303716 | -0.0944        | 0.9974           |
| 1:104864464 | -0.0837        | 1.0000           |

**Figure 8.** Individual signal intensity obtained by PennCNV

#### Rule CREATE PennCNV INPUT FILES

**Description:** Generate the Population frequency of B allele (PFB) file and the GCModel file used by PennCNV in the CNV detection.

**Input:** a file with the list of names for each intensity file created, and the SNPs list file (name, Chr, Position).

**Output:** the pfb file and the gc-model file.

A log file is generated inside the logs directory with the module name (logs/data\_conversion.log) showing useful information for future reports.

## 2. Module Data Calling

### Rule DETECT\_CNVS

**Description:** Run the PennCNV for CNVs detection

**Input:** the pfb file and the gc-model file.

**Output:** list of calls per individual in PennCNV \*.rawcn (Figure 9) and log files. This file shows CNV chromosome coordinates, number of SNPs contained in the CNV, length, type of variation (CN=0 or 1 means there is a deletion and CN>=3 means there is a duplication), the starting marker identifier and the ending marker identifier in the CNV, respectively.

A log file is generated inside the logs directory with the module name (logs/data\_calling.log) showing useful information for future reports.

```
chr1:149095346-149143879      numsn=4      length=48,534      state2,cn=1 split1 startsn=GSA-1:149095346 endsn=GSA-rs79760750 conf=18.072
chr4:156963692-156966696      numsn=3      length=3,005      state2,cn=1 split1 startsn=rs4691238 endsn=GSA-rs112975832 conf=12.029
chr5:97066021-97096042       numsn=5      length=30,022      state2,cn=1 split1 startsn=GSA-rs76728709 endsn=rs4380692 conf=20.359
chr6:76261562-76265642       numsn=3      length=4,081      state2,cn=1 split1 startsn=rs61234544 endsn=rs9360921 conf=16.661
chr11:95979334-95980389      numsn=3      length=1,056      state2,cn=1 split1 startsn=rs79686966 endsn=rs535912 conf=14.694
chr14:106208082-106232585    numsn=3      length=24,504      state2,cn=1 split1 startsn=GSA-rs11621259 endsn=B0T2-rs10136766 conf=8.561
chr22:42527793-42528976      numsn=4      length=1,184      state2,cn=1 split1 startsn=rs1080989 endsn=rs28360521 conf=17.310
chr3:155481492-155505991     numsn=6      length=24,500      state5,cn=3 split1 startsn=chr3-155481492 endsn=rs112074828 conf=16.732
chr7:141763387-141775383     numsn=10     length=11,997      state5,cn=3 split1 startsn=chr7-141763387 endsn=GSA-rs77848363 conf=28.198
chr8:12009572-12009597      numsn=3      length=26          state5,cn=3 split1 startsn=GSA-rs9773610 endsn=GSA-rs75619199 conf=10.326
```

**Figure 9.** PennCNV calling file rawcn format.

## 3. Module Data Clean

### Rule FILTER\_LOW\_QUALITY\_SAMPLES

**Description:** filter low quality samples based mainly on LRR and BAF statistics.

**Input:** rawcn and log file from PennCNV.

**Output:** rawcn file with good quality samples and their CNVs, file with the list of samples passed the QC, and summary file with all samples and their calls (passed and not passed the QC) include LRR and BAF statistics.

A plot showing the statistics LRR, BAF and WF distribution is generated. Also a log file is created inside the logs directory with the module name (logs/data\_clean.log) showing useful information for future reports.

### Rule REMOVE\_SPURIOUS\_CNVS

**Description:** Remove CNVs from spurious regions such as HLA, centromere and telomere.

**Input:** rawcn file with good quality samples and their CNVs.

**Output:** rawcn file with good quality samples and their CNVs not located on spurious regions  
A log file is generated inside the logs directory with the module name (logs/data\_clean.log) showing useful information for future reports.

### Rule MERGING\_ADJACENT\_CNVS

**Description:** Merge adjacent CNVcalls, in one single one, that could have been split from a large CNVs (>500kb) into smaller parts.

**Input:** rawcn file with good quality samples and their CNVs not located on spurious regions.

**Output:** merged CNVs in rawcn format. Violin plot showing the calls number distribution before and after the merging process is also generated.

A log file is generated inside the logs directory with the module name (logs/data\_clean.log) showing useful information for future reports.

## **Rare copy number variants (CNV) pipeline**

### **1. Module Data Conversion**

#### Rule CONVERT\_PENNCNV\_TO\_PLINK\_FORMAT

**Description:** convert rawcn files from PennCNV to Plink cnv, fam and map files. File with **all** CNVs previously detected at Module Data Calling in QC pipeline is converted in plink format. Information about number of calls, number of cases, controls, males and females can be extracted from these files. External function (function.sh) is referenced in this rule to execute external bash code.

**Input:** file with all calls (rawcn format) previously detected.

**Output:** cnv, fam and map files for all calls (Plink format).

A log file is created inside the logs directory with the module name (logs/data\_conversion.log) showing useful information for future reports.

#### Rule EXTRACT\_CORE\_SAMPLES

**Description:** extract CORE homogeneous unrelated samples, and their CNVs in Plink format files. Users should provide the ID list of unrelated samples (e.g. European descendant individuals). This rule could be modified according to the project requirements. External bash code, included in the function.sh file, is called in this rule.

**Input:** rawcn file with clean CNVs (passed QC and merged CNVs) and the list of unrelated European samples (IDs in a column).

**Output:** core samples and clean CNVs in cnv, fam and map plink files format.

A log file is created inside the logs directory with the module name (logs/data\_conversion.log) showing useful information for future reports.

#### Rule EXTRACT\_FINAL\_CNVS

**Description:** A set of clean big calls covered by a minimum of markers is extracted in this step. Both, length and number of markers, can be set at variables.py file (by default we use > 50 kb and > 5 markers). This rule could be adapted or removed according to the project requirements.

**Input:** rawcn file with core samples and clean calls (passed QC and merged CNVs)

**Output:** core samples and clean CNVs (passed QC and merged CNVs) bigger than 50 kb and covered by more than 5 markers, in cnv, fam and map plink files format. Individuals with 0 CNVs are also included in this final set.

A log file is created inside the logs directory with the module name (logs/data\_conversion.log) showing useful information for future reports.

## 2. Module Burden Analysis

### Rule BURDEN CNVs

**Description:** Perform the global burden test for CNVs in the core samples, cases vs. controls.

**Input:** CNVs for core samples in cnv, fam and cnv.map files.

**Output:** burden test result for CNVs in the core samples, showed at \*.summary.mperm and grp.summary files (Figure 10).

A log file is created inside the logs directory with the module name (logs/burden\_analysis.log) showing useful information for future reports.

```
*.cnv.grp.summary
TEST      GRP      AFF      UNAFF
N          ALL      2270     7728
RATE      ALL      1.92     1.927
PROP      ALL      0.8477   0.8536
TOTKB     ALL      362.6    356.8
AVGKB     ALL      160.5    157.2

*.cnv.summary.mperm
CHR      SNP      EMP1
S        RATE   0.562744
S        PROP   0.706829
S        KBTOT  0.341366
S        KBAVG  0.284572
S        GRATE   1
S        GPROP   1
S        GRICH   1
```

**Figure 10.** Burden test output. This test reports four tests in both cases and controls, RATE (Number of segments), PROP (Proportion of sample with one or more segments), TOTKB (Total kb length spanned), and AVGKB (Average segment size). Tests are based (1-sided) on comparing these metrics in cases versus. For more detail see [https://zzz.bwh.harvard.edu/plink/cnv.shtml#write\\_cnvlist](https://zzz.bwh.harvard.edu/plink/cnv.shtml#write_cnvlist). In this example there is not apparent significant difference in any of the four test (EMP1 > 0.05)

### Rule SPLIT CNVs DEL AND DUP

**Description:** Split CNVs in deletions and duplications and then obtain the frequency of deletions and duplications in cases vs. controls.

**Input:** CNVs for core samples in cnv, fam and cnv.map files.

**Output:** \*.cnv.indiv files for deletions and duplications in core samples. Also generate a tsv file containing the deletions and duplications ratio, split by length (by default 50kb, 100kb, 200kb, 500kb and 1Mb) which can be configured in variable.py file. Moreover, generate a tsv file containing the total mean length (in KB) for deletions and duplications in cases and

controls (Figure 11); and a plot showing the CNVs length-distribution in cases vs. controls (Figure 12).

A log file is created inside the logs directory with the module name (logs/burden\_analysis.log) showing useful information for future reports.

| Deletion frequency TSV file |        |        |            |
|-----------------------------|--------|--------|------------|
| pheno                       | length | numCNV | ratio      |
| 1                           | 50KB   | 3502   | 0.873317   |
| 1                           | 100KB  | 1667   | 0.415711   |
| 1                           | 200KB  | 424    | 0.105736   |
| 1                           | 500KB  | 83     | 0.0206983  |
| 1                           | 1000KB | 12     | 0.00299252 |
| 2                           | 50KB   | 1027   | 0.868866   |
| 2                           | 100KB  | 460    | 0.389171   |
| 2                           | 200KB  | 132    | 0.111675   |
| 2                           | 500KB  | 30     | 0.0253807  |
| 2                           | 1000KB | 13     | 0.0109983  |

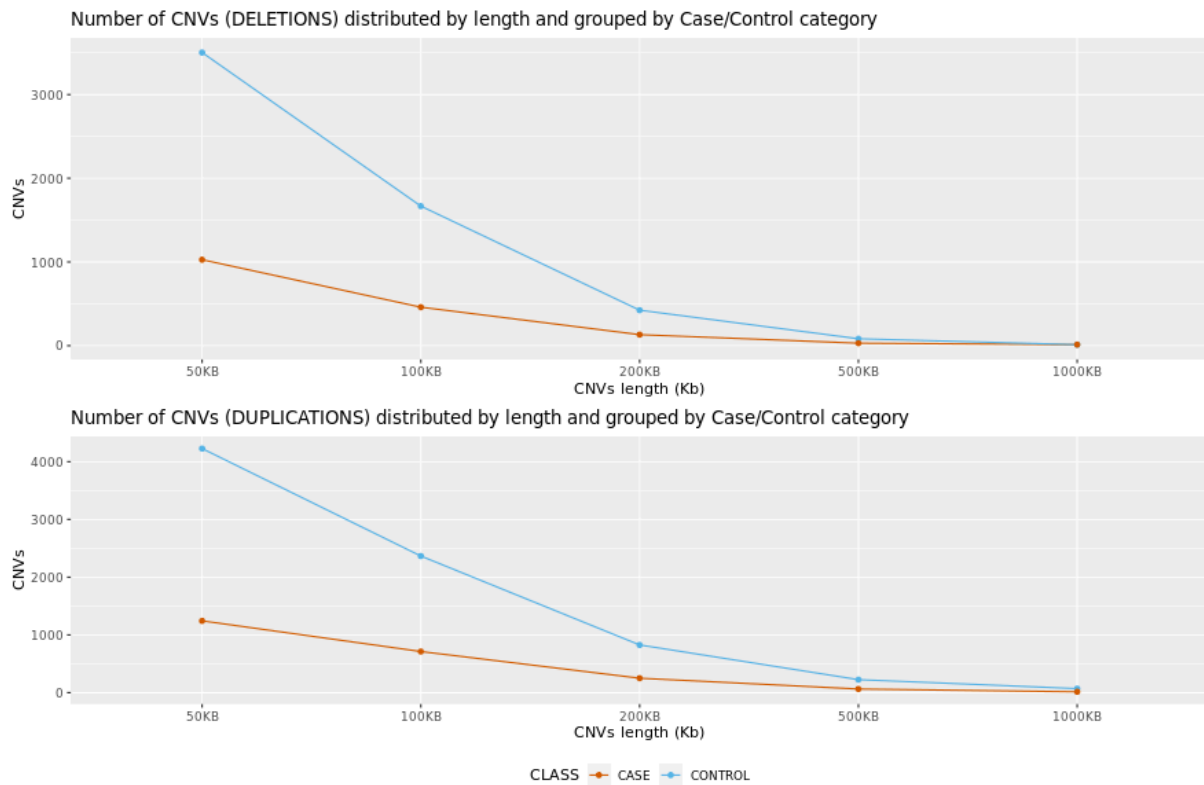
  

| Total mean length (in KB) TSV file |                     |          |                          |
|------------------------------------|---------------------|----------|--------------------------|
| pheno                              | tot_length_CNVs(Kb) | tot_CNVs | tot_mean_length_CNVs(Kb) |
| 1                                  | 463500              | 3502     | 132.353                  |
| 2                                  | 145182              | 1027     | 141.365                  |

| Duplications |                     |          |                          |
|--------------|---------------------|----------|--------------------------|
| pheno        | tot_length_CNVs(Kb) | tot_CNVs | tot_mean_length_CNVs(Kb) |
| 1            | 757743              | 4226     | 179.305                  |
| 2            | 218126              | 1243     | 175.484                  |

**Figure 11.** TSV files generated in the module Burden Analysis. The first example shows the deletions frequency classified by length and phenotype (1:controls, 2:cases); and the second example shows the mean length for all deletions and duplications in controls and cases, 1 and 2 for pheno column respectively.



**Figure 12.** Length distribution in deletions and duplications. In this example CNVs are split in deletions and duplications and classified by cases and controls. Then they are distributed in 5 length intervals [50kb, 100kb, 200kb, 500kb, 1000kb]. This plot shows that bigger is the CNV length less is the number of CNVs in that interval, in cases and controls.

### 3. Module Rare CNVs Analysis

#### Rule GET\_REF\_CONTROLS\_COMMON\_CNVS

**Description:** Based on a subset of control samples, it obtains the **common** CNVs (deletions and duplications). The number of controls included in this subset can be defined in the config file `variables.py`. Common CNVs are defined as CNVs with high frequency, and this frequency also can be defined in `variables.py`.

**Input:** The ID list for the set of reference controls. These samples should be obtained randomly and including the same number of males and females (and any other population variable as nationality).

**Output:** \*.cnv, \*.fam and \*.cnv.map files for reference controls and **common variants** (high frequency) split by type: deletions and duplications

A log file is created inside the logs directory with the module name (`logs/rare_cnvs.log`) showing useful information for future reports.

#### Rule FILTER\_REFERENCE\_CONTROLS

**Description:** Remove the reference control samples and their CNVs, from the original control cohort, since this dataset has been used to extract the common CNVs.

**Input:** Reference controls ID list, \*.cnv, \*.fam and \*.cnv.map for common variants and reference controls.

**Output:** \*.cnv, \*.fam and \*.cnv.map files for the new set of samples and CNVs which exclude the reference controls.

A log file is created inside the logs directory with the module name (logs/rare\_cnvs.log) showing useful information for future reports.

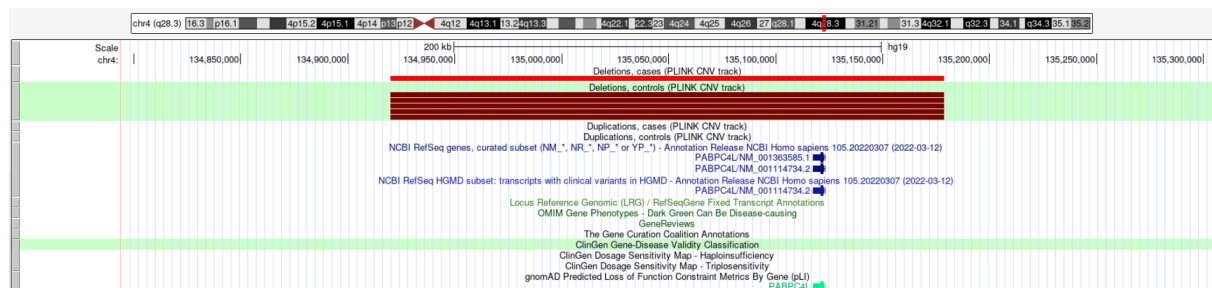
### Rule GET\_RARE\_CNVS

**Description:** Extract rare CNVs, deletions and duplications. All CNVs overlapping at least 50% with common variants will be removed to retain just low frequency CNVs.

**Input:** \*.cnv, \*.fam, \*.map files for common variants and new dataset excluding reference controls.

**Output:** \*.cnv, \*.fam and \*.cnv.map files for rare deletions and duplications. Also BED files were created for a graphic visualization in the Genome Browser and any genome viewer as IGV (Figure 13)

A log file is created inside the logs directory with the module name (logs/rare\_cnvs.log) showing useful information for future reports.



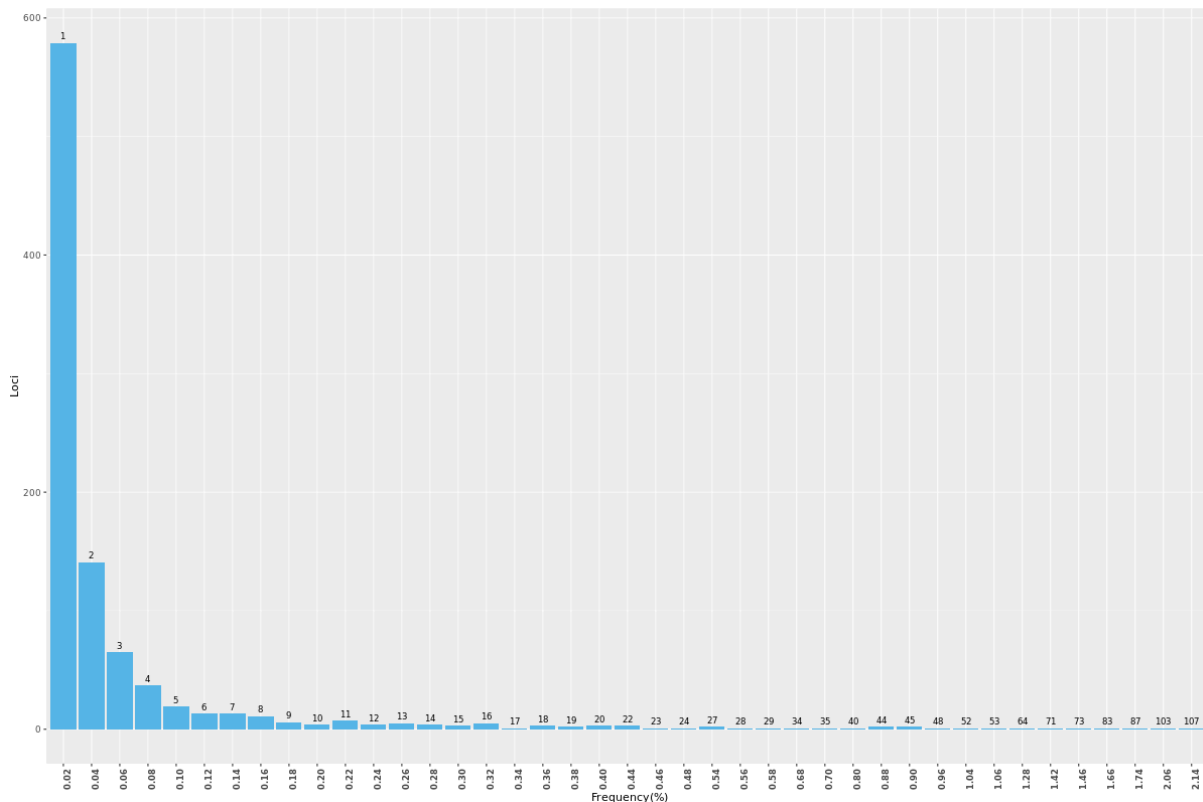
**Figure 13.** CNVs visualized in the Genome Browser. Deletions in cases and controls (light red versus dark red) visualized using the BED files.

### Rule FREQUENCY\_CNVS\_INSPECTION

**Description:** Inspect the frequency of rare CNVs, deletions and duplications. CNVs regions (CNVR) are defined to calculate the CNVs frequencies. After that, frequency is plotted to verify that frequencies are approximately below the frequency threshold set in the variable.py file (Figure 14).

**Input:** \*.cnv, \*.fam, \*.map files for rare deletions and duplications, and the fam file for all CNVs included in the study to calculate the frequency in each CNVR.

**Output:** \*.cnv, \*.fam and \*.cnv.map files for CNVR. Bar plot showing the CNVs frequency distribution.



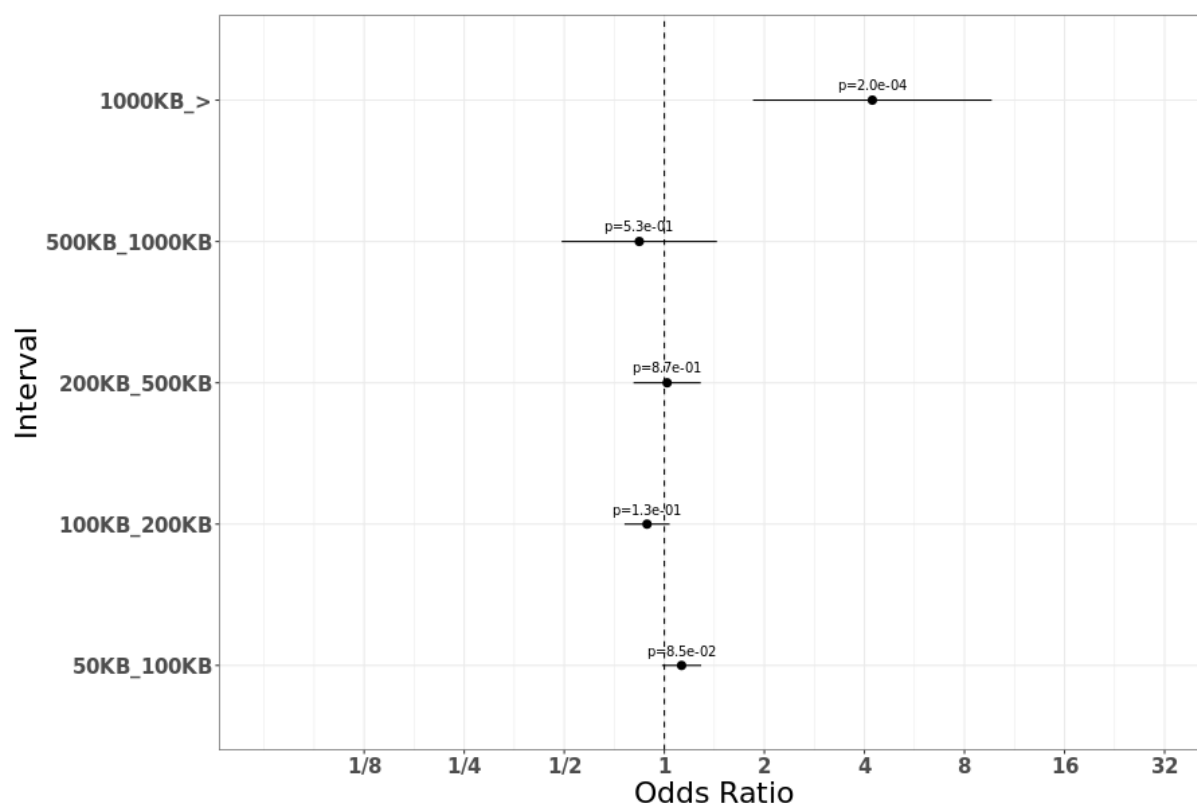
**Figure 14.** Deletions frequency distribution. After removing common variants, rare CNVs frequency are inspected showing the bar plot with the CNVs frequency which should be approximately below the threshold defined in the config file (in this example frequency should be approximately less than 2%) . The Y axis shows the number of CNVs in a specific locus (region), the X axis shows the CNVs frequency, and the number in the bar shows the amount of CNVs which represent this frequency. E.g for the second bar, there are 141 regions with 2 deletions in each, and the frequency of these deletions is equal to 0.04%, in a total of 4992 samples (2/4992).

### Rule FREQUENCY CNVs ASSESSMENT

**Description:** assess the CNVs frequencies. A frequencies summary file for all deletions and duplications in cases and controls is created. This file includes the two proportion test evaluation and the odds ratio (OR) estimation. Similarly, a summary file containing the deletions and duplications frequencies split by CNVs interval sizes, and the statistics tests is created in this step. These results were graphically plotted in two forest plots, showing the frequencies confidence interval in each CNVs interval size, and the p-value associated (Figure 15). Also, bars plots are generated showing the proportion of individuals with specific number of CNVs, and the proportion of individuals with CNVs in each size interval.

**Input:** rare CNVs (deletions and duplications) \*.cnv, \*.fam and \*.cnv.map files.

**Output:** summary frequencies \*.csv files, forest plot and samples distribution bar plots. A log file is created inside the logs directory with the module name (logs/rare\_cnvs.log) showing useful information for future reports.



**Figure 15.** Deletions frequencies forest plot.

#### 4. Module Rare CNVs Enrichment

##### Rule GENIC ENRICHMENT ANALYSIS

**Description:** Perform the basic geneset-enrichment test to evaluate which genes are enriched for CNVs. This test is a permutation-based test with 10000 null permutations to generate empirical p-values. Also, Gene regions borders are set at 20 kb around each region. Both, permutations and regions border, can be adjusted inside the rule.

**Input:** rare deletions and duplications \*.cnv, \*.fam and \*.cnv.map files, and the gene annotation for the entire genome.

**Output:** \*.cnv.burden files with the Performing GLM-based CNV burden test results. Also, a text file containing the list of genes (chromosome, coordinates, number of CNVs in cases and controls) overlapped by any CNV is generated.

A log file is created inside the logs directory with the module name (logs/enrichment\_rare\_cnvs.log) showing useful information for future reports.

## Rule PATHWAY\_ENRICHMENT\_ANALYSIS

**Description:** Perform the basic geneset-enrichment test to evaluate whether a subset of genes are enriched, relative to the whole genome. This test is a permutation-based test with 10000 null permutations to generate empirical p-values. Also, Gene regions borders are set at 20 kb around each region. Both, permutations and regions border, can be adjusted inside the rule.

**Input:** rare deletions and duplications \*.cnv, \*.fam and \*.cnv.map files, and the gene annotation for the entire genome and file of gene names forming the pathway to be tested.

**Output:** \*.cnv.burden files with the Performing GLM-based CNV burden test results. Also, a text file containing the list of genes (chromosome, coordinates, number of CNVs in cases and controls) overlapped by any CNV is generated.

A log file is created inside the logs directory with the module name (logs/enrichment\_rare\_cnvs.log) showing useful information for future reports.

## **Input Files Specification**

### **Illumina SNP array**

The calling and QC pipeline requires as an input the Illumina report (see Figure 7) to extract the signal intensity files (see Figure 8). Users may get this file directly from the genotyping center, or they may get a GenomeStudio project containing the signal intensity values (Log R Ratio and B Allele Frequency) for all markers in all samples in a given project. Then, the illumina report can be extracted using the GenomeStudio software following the instructions provided in the [GenomeStudio Genotyping Module v2.0 User Guide](#), Generate a Final Report section.

Alternatively, users may get the genotyping data in BeadStudio project file format. In this case the Illumina BeadStudio software has to be used to extract the signaling intensity files. PennCNV provides a guide to extract these files in the section [Preparing input signal intensity files from BeadStudio project files](#).

The second input file required by the calling and QC pipeline is the SNPs table file (see Figure 6) used to extract the position of each SNP (chr, bp1, bp2). Users may get this file from the genotyping center or it can be also extracted from the original GenomeStudio/Bedasuit project.

Rule CREATE\_INTENSITY\_FILES within the module Data-Conversion contains the code to process both files and obtain the next input files to execute PennCNV. Users can modify or even discard this rule to adapt their own data according to the format files required as we explain below.

## Affymetrix SNP array

For genotyping data from Affymetrix platforms, raw CEL files should be converted to obtain the signal intensity files and the SNPs position file. Rule CREATE\_INTENSITY\_FILES can be modified to include the new code, or users can prepare the appropriate input file formats a priori and discard this rule.

Procedure to convert Affymetrix genotyping data is very well detailed in the [PennCNV-Affy tutorial](#). PennCNV especially supports Genome-wide human array, Mapping array, Axiom array and Cytoscan HD array.

We will provide an example of how to adapt the module “Data Conversion”, in calling and QC pipeline, to generate the signals files from Affymetrix Genome-Wide Human SNP 6.0 Array.

### Requirements for Genome-wide 6.0 array analysis (adapted from [PennCNV-Affy tutorial](#))

1. PennCNV-Affy programs and library files (gw6.tar.gz):
  - a. affygw6.hg18.gcmodel
  - b. affygw6.hg18.pfb
  - c. affygw6.hmm
  - d. hapmap.quant-norm.normalization-target.txt
  - e. generate\_affy\_geno\_cluster.pl
  - f. normalize\_affy\_geno\_cluster.pl
2. Libraries from Genome-wide 6.0 array [library files from ThermoFisher](#) (genomewidesnp6\_libraryfile.zip):
  - a. GenomeWideSNP\_6.cdf
  - b. GenomeWideSNP\_6.birdseed.models
  - c. GenomeWideSNP\_6.specialSNPs
3. Libraries from [Affymetrix Power Tools \(APT\) software package](#) (apt\_2.12.0\_linux\_64\_x86\_binaries.zip):
  - a. apt-probeset-genotype
  - b. apt-probeset-summarize

**NOTE:** These libraries and programs were collected and are included in the [Affymetrix.tar.gz file](#).

### Changes in Data Conversion module

1. Download [Affymetrix.tar.gz file](#) and uncompress it into qc-cnv/lib
2. Create the text file “listfile” containing the list of CEL files:
  - a. echo “cel\_files” > listfile
  - b. for i in `realpath \*.CEL`; do echo \$i; done >> listfile

- c. The path containing this file will be added in the config.json file (see 4.)

**NOTE:** The absolute path should be contained in `listfile`

```
cel_files
/absolute_path_to/CEL/GSM337641.CEL
/absolute_path_to/CEL/GSM337642.CEL
/absolute_path_to/CEL/GSM337643.CEL
/absolute_path_to/CEL/GSM337644.CEL
/absolute_path_to/CEL/GSM337645.CEL
/absolute_path_to/CEL/GSM337646.CEL
/absolute_path_to/CEL/GSM337647.CEL
/absolute_path_to/CEL/GSM337648.CEL
/absolute_path_to/CEL/GSM337649.CEL
```

3. Replace `module1-data-conversion.snake` content with `module1-data-conversion.Affymetrix.snake` included in `qc-cnv/qc-pipeline/snakefiles` directory.

4. Add the two lines (in bold) the input statement in `module2-data-calling.snake`

```
rule DETECT_CNVs:
  input:
    pbffile = config['pfb_file'],
    gcmodelfile = config['gcmodel_file'],
    listfile = config['list_signal_files_file'],
    snpfile = config['snp_file']
```

5. Modify `configure.js` file:

- Include the path to `listfile`
- Unlike Illumina arrays, hmm, pfb and gcmodel files are not generated when Affymetrix array data is used. These files are provided by PennCNV in the PennCNV-Affy (gw6.tar.gz) and they are included in the Affymetrix.tar.gz file.

```
"list_CEL_file": "/path_to/CEL/listfile"
"hmm_file": "qc-cnv/lib/Affymetrix/affygw6.hmm",
"pfb_file": "qc-cnv/lib/Affymetrix/affygw6.hg18.pfb",
"gcmodel_file": "qc-cnv/lib/Affymetrix/affygw6.hg18.gcmodel",
```

#### NOTES:

- Affymetrix Genome-Wide Human SNP 6.0 Array data from [GEO \(GSE13372\\_RAW.tar\)](#) was downloaded to test the calling and QC pipeline.
- As is mentioned in PennCNV guide, Affymetrix data contains more noise than Illumina data, therefore a more liberal QC threshold should be used.
- To test data in (1), LRR\_SD (qclrrsd) and NumCNVs (qcnumcnv) parameters were adjusted (0.4 and 200, respectively) in `variables.py`.